

4.11 Writing Equations of Perpendicular Lines

Lesson Introduction

 Benchmark	MA.912.AR.2.3 Write a linear two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.		 Learning Targets	- I can write the equation of a line that is perpendicular to a given line through a given point. - I can write the equation of a line that is perpendicular to a given line when given two points. - I can write the equation of a line that is perpendicular to a given line when given a point and the slope.	
 Benchmark Coherence	Building On		Working Toward		
	MA.8.AR.4.3 Given a mathematical or real-world context, solve systems of two linear equations by graphing.		MA.912.GR.3.2 Given a mathematical or real-world context, use coordinate geometry to classify or justify definitions, properties and theorems involving circles, triangles, or quadrilaterals.		
 Essential Questions	- What is the relationship between two lines that are perpendicular to one another? - Which key features play a role in determining if two equations represent lines that are perpendicular to one another? - How would you write the equation of a line that is perpendicular to another line?				
 Lesson Narrative	Like Lesson 10, this lesson guides students through a discovery of the mathematical relationship between perpendicular lines. Students formalize that relationship as “opposite and reciprocal” before applying it in the context of vertical and horizontal lines.				
 Component Summary	4.11.1 Warm-Up (~10 min.)	Spiraled practice for writing the equations of parallel lines (<i>SE p. 220</i>)	4.11.3 Guided Practice (10 min.)	Students apply their newly formalized understanding to perpendicular lines (<i>SE p. 222</i>)	
	4.11.2 Guided Instruction (15 min.)	Students explore the mathematical relationship of perpendicular lines before formalizing their definition (<i>SE p. 221</i>)	4.11.4 Your Turn! (10 min.)	Independent practice for writing the equations of perpendicular lines (<i>SE p. 223</i>)	
	4.11.5 Wrap-Up (~5 min.)	Learning Pyramid self-reflection on learning progress regarding perpendicular lines (<i>SE p. 224</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Perpendicular <u>Additional Terms:</u> - Reciprocal		N/A		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may mix up the relationship of slope with parallel and perpendicular lines. - Students may believe “reciprocal” refers to the opposite sign (i.e., the reciprocal of 5 is -5).	Lesson 12 provides more practice for writing the equations of perpendicular lines. Help students embrace the struggle and utilize the tools of growth mindset and self-reflection to be effortful in their learning. Prioritize conversation and process as students make sense of information before more practice in Lesson 12.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 4.11.2 Guided Instruction question 2d positions students to begin making connections between their informal observations of perpendicular lines and their formal mathematical properties. Utilize a Think-Pair-Share routine to provide space and opportunity for multiple attempts to facilitate more confident, precise responses.	MA.K12.MTR.1.1: Participate in Effortful Learning Both the 4.11.2 Guided Instruction and 4.11.3 Guided Practice position students to discover learning rather than being “told” what they’re learning. Consider recognizing and praising student actions like asking questions, making connections, and taking risks during the learning process.
 Differentiate Instruction	Some students might “just want the steps” when writing the equation of a line. Questions like 2d in 4.11.2 Guided Instruction provide an intentional opportunity for students to make sense of the information rather than just memorizing shortcuts or steps. Use opportunities like this to provide additional entry points for some students while extending content for others.	
	Consider your personal classroom strategies for vocabulary and language development for terms like “perpendicular” or “reciprocal” using strategies like Frayer Model, Vocab Flash Cards, Word Walls, and more.	
Lesson Notes:		

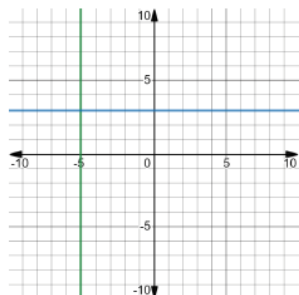
4.11.1 Warm-Up: Horizontal and Vertical Lines

Component Overview: Have students complete independently. This provides additional practice with writing the equation of a line using parallel, horizontal, and vertical lines.



4.11.1 Warm-Up: Horizontal and Vertical Lines

The lines $x = -5$ and $y = 3$ are shown on the coordinate grid.



Rodrigo completed the table for the lines $x = -5$ and $y = 3$.

Line	Slope	Explanation
$x = -5$	$\frac{2}{0}$	If I start at the point $(-5, 0)$, I can go up 2 and still be on the line. In terms of $\frac{\text{rise}}{\text{run}}$, I rise 2 and run 0.
$y = 3$	$\frac{0}{2}$	If I start at the point $(0, 3)$, I can go up 0 and over 2 and still be on the line. In terms of $\frac{\text{rise}}{\text{run}}$, I rise 0 and run 2.

1. Give another slope Rodrigo could have used for each line.

Line	Slope
$x = -5$	Answers will vary. $\frac{17}{0}$
$y = 3$	Answers will vary. $\frac{0}{11}$

2. Write the equation of the line that passes through $(2, 1)$ and is parallel to $x = -5$.
 $x = 2$
3. Write the equation of the line that passes through $(2, 1)$ and is parallel to $y = 3$.
 $y = 1$
4. What is the point where $x = -5$ and $y = 3$ intersect?
 $(-5, 3)$



Avoid using language like “opposite” or “reciprocal” during 4.11.1 Warm-Up. The next activity includes question prompts to begin facilitating that connection for students, which can then be connected back to 4.11.1 Warm-Up. Instead, utilize 4.11.1 Warm-Up as an additional opportunity for spiraling content from Lessons 9-10 for a strong foundation for this lesson.

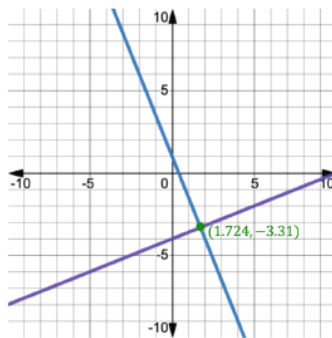
4.11.2 Guided Instruction: Writing Equations of Perpendicular Lines Through a Point

Component Overview: Students learn how to write the equation of a line that is perpendicular to a line through a point not on the line.



4.11.2 Guided Instruction: Writing Equations of Perpendicular Lines Through a Point

- In the warm-up, what is the relationship between the lines $x = -5$ and $y = 3$?
Answers will vary. $x = -5$ is a vertical line. $y = 3$ is a horizontal line. They are perpendicular lines.
- Two lines are graphed on the coordinate grid. The lines are perpendicular to each other.



- Find the slope of each line.

Line	Slope
Purple	$\frac{2}{5}$
Blue	$-\frac{5}{2}$

- Write the equation of the purple line.
 $y = \frac{2}{5}x - 4$
- Write the equation of the blue line.
 $y = -\frac{5}{2}x + 1$



This activity begins with a purposeful connection to 4.11.1 Warm-Up. After students have completed 4.11.1 Warm-Up, immediately jump into question 1 in 4.11.2 Guided Instruction to begin facilitating those connections. Students will then begin discovering the relationship between the slopes of perpendicular lines in questions 2a-c before formalizing their understanding in question 2d.

What do you notice about the point of intersection in 4.11.1 Warm-Up? (It looks like a right angle.) Does the point of intersection of the two lines in question 2 have the same relationship? (Students may believe lines can only be perpendicular if they are “straight” or vertical like in 4.11.1 Warm-Up).



Students will begin making connections to the slope of perpendicular lines before formalizing the language in question 2d below.

What do you notice about the relationship between the slopes of the blue and purple line? Was this also true for 4.11.1 Warm-Up example? Do you believe this will always be true? Why or why not?

4.11.2 Guided Instruction: Writing Equations of Perpendicular Lines Through a Point (continued)

- d. Discuss with your partner how to write the equation of a line that is perpendicular to another line. Write notes from the discussion.

Answers will vary. First, I find the slope of one line. The slope of a line perpendicular to it will be the opposite reciprocal. Then, I use the opposite reciprocal for the slope and a point on the new line to write the equation in point-slope form.



Lines are **perpendicular** if the slopes are opposites and reciprocals.

3. Write the equation of the line, in all three forms, that is perpendicular to the line that passes through the points $(-5, -2)$ and $(3, 2)$ and goes through the point $(4, -1)$.

Point-Slope Form	Slope-Intercept Form	Standard Form
$y + 1 = -2(x - 4)$	$y = -2x + 7$	$2x + y = 7$



Think-Pair-Share

Consider providing multiple entry-points to students who are making sense of information using a Think-Pair-Share routine. Think-time gives students a chance for internal processing, pairing students gives space for verbal processing and auditory learners, and facilitating a class share-out gives opportunity for refining more precise answers to question 2d.



Does your classroom utilize any vocabulary strategies like a Word Wall, Anchor Charts, or similar tools? Consider using this opportunity to integrate “perpendicular” into students’ schema of lines using one of those strategies.



What does “opposites and reciprocals” mean?

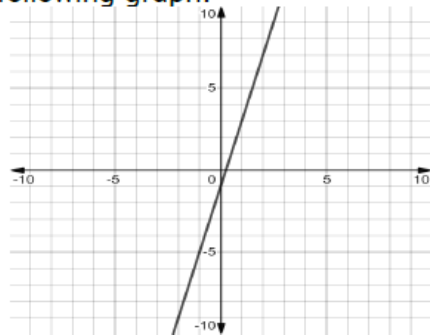
4.11.3 Guided Practice: Writing Equations of Perpendicular Lines Through a Point

Component Overview: Students write the equation of a line that is perpendicular to a line through a point not on the line in various forms.



4.11.3 Guided Practice: Writing Equations of Perpendicular Lines Through a Point

1. Consider the following graph.



- Write the equation of the line that is perpendicular to the line shown on the coordinate grid and passes through the point (5,2).
 $y - 2 = -\frac{1}{4}(x - 5)$ or any equivalent equation
- What is the point where the two lines intersect?
(1,3)
- Write the equation of the line, in all three forms, that is perpendicular to the line $25x - 5y = 45$ and goes through the point (0,7).

Point-Slope Form	Slope-Intercept Form	Standard Form
$y - 7 = -\frac{1}{5}(x - 0)$ OR $y - 7 = -\frac{1}{5}x$	$y = -\frac{1}{5}x + 7$	$\frac{1}{5}x + y = 7$ OR $x + 5y = 35$

2. Write the equation of the line, in all three forms, that is perpendicular to the line $y - 8 = \frac{5}{6}(x + 11)$ and goes through the point, (-10, -3).

Point-Slope Form	Slope-Intercept Form	Standard Form
$y + 3 = -\frac{6}{5}(x + 10)$	$y = -\frac{6}{5}x - 15$	$\frac{6}{5}x + y = -15$



Allow informal observation throughout the first two activities to dictate how “guided” 4.11.3 Guided Practice should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.



Students may conceptually grasp the concept of perpendicular lines but struggle with the calculations and mathematical relationship. Consider what specific skills or entry points might need additional small group instruction based on your observations and data.

- Identifying slope from a graph.
- Identifying an opposite reciprocal.
- Converting between forms, particularly standard to slope-intercept, to find slope.
- Number sense and calculating/simplifying.



For students who finish early, challenge them to answer the following prompts:

- How is the mathematical relationship between perpendicular lines the same or different than parallel lines?
- Is it possible to have a set of perpendicular lines that are also parallel to a third line? Give an example or counterexample.

4.11.4 Your Turn!

Component Overview: Students work independently to write the equation of a line that is perpendicular to a line through a point.



4.11.4 Your Turn!

1. Write the equation of the line that goes through the point $(6, -8)$ and is perpendicular to the line that passes through the points $(-7, -12)$ and $(-7, 6)$.

$$y = -8$$

2. Write the equation of the line that goes through the point $(2, 3)$ and is perpendicular to the line

$$-8x - 6y = -48.$$

$$y = \frac{3}{4}x + \frac{3}{2} \text{ or any equivalent equation}$$



Students have been built up to now attempt to write the equation of a line through a point not on a line on their own. While students are working, consider collecting observational data using a rubric to inform your instruction for Lesson 12 and to determine which students may need support using a small group. Pair your observations with the student reflection in 4.11.5 Wrap-Up.

4.11.5 Wrap-Up: Reflection

Component Overview: This Learning Pyramid provides an opportunity for students to provide qualitative data regarding their learning progression.



4.11.5 Wrap-Up: Reflection

1. Complete the learning pyramid.

Answers will vary.

One question I would like to have answered ...	
One mistake I learned from today ...	One thing I am not sure about ... YET!
One thing I am confident I know is ...	



For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. Students could also use sentence stems. All forms can show an equal level of rigor and conceptual understanding.